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Maximizing Value from Drill Cuttings. A Case Study in the Eagle Ford Using Robotic XRF Logging

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Abstract

Drill cuttings, often undervalued in traditional workflows, represent a readily available byproduct of drilling operations with substantial untapped potential. While conventionally used for rudimentary lithological descriptions during mudlogging, recent technological advancements have transformed drill cuttings into a viable, real-time source of high-resolution geological and geochemical data. This case study from the Eagle Ford Shale in the structurally complex Southwest–Northeast Texas Trend demonstrates the development and field-scale deployment of an innovative, automated workflow. The approach leverages robotic sampling technologies, advanced geochemical instrumentation, and data integration frameworks to extract maximum subsurface insight from what was once considered drilling waste.

The Eagle Ford's geologic setting—marked by strong lateral facies variability, abrupt mechanical changes, and sub-seismic scale faulting—presents a challenging environment for traditional downhole tools such as Measurement While Drilling (MWD), gamma-ray (GR) sensors, and even some Logging While Drilling (LWD) devices. These tools are frequently limited by signal noise, tool vibration, sensor failure, and resolution constraints, particularly at high penetration rates. In response, Diversified developed and implemented a fully automated drill cuttings acquisition and analysis system. The system integrates high-frequency robotic sampling with portable X-ray fluorescence (XRF) and Laser-Induced Breakdown Spectroscopy (LIBS), enabling rapid in situ measurement of major oxides and trace elements in near real time. The workflow includes automatic data flagging, built-in QA/QC controls, and robust calibration routines.

In this system, surrogate gamma-ray profiles are generated from elemental concentrations of potassium (K), thorium (Th), and uranium (U), which are converted into American Petroleum Institute (API) units using regression-based models. These surface-derived gamma-ray curves offer a cost-effective and high-resolution alternative to conventional GR logging tools. Furthermore, the introduction of robotics not only standardizes and accelerates sampling but also minimizes human exposure to hazardous rigsite conditions, including those involving H₂S gas, high-pressure mud systems, and rotating equipment.

Across an 80-well dataset collected over two years, the integration of robotic sampling and surface-based elemental analysis improved subsurface resolution and enhanced operational decision-making. The ability to collect and analyze samples at 5-foot intervals significantly outperforms manual sampling,

which typically occurs every 100 feet due to operational limitations. The increased vertical resolution of geochemical data allowed operators to refine landing zones, avoid undesirable lithologies, and steer laterals within optimal mechanical and geochemical windows. In multiple instances, the system provided critical data during downhole tool failure events, enabling uninterrupted drilling operations.

This study demonstrates that combining automation with advanced surface analytics offers a reliable, scalable, and economically viable alternative to conventional formation evaluation tools. The success of the approach underscores the value of repurposing drill cuttings as a primary dataset and sets a precedent for their application in other unconventional reservoirs worldwide.

Keywords: automation, robotics, XRF, LIBS, surface gamma ray, Eagle Ford Shale, chemostratigraphy, high-resolution cuttings analysis, real-time drilling optimization

Introduction

Since the early days of petroleum exploration, drill cuttings have served as the primary physical evidence of formations penetrated by the drill bit. Captured and evaluated by mud loggers, these fragments of rock traditionally provide lithological descriptions, indications of hydrocarbon shows, and basic stratigraphic control. However, the inherent limitations of manual cuttings collection—particularly with respect to resolution, sample representativeness, and turnaround time—have historically relegated cuttings to a supporting role. In modern horizontal drilling operations, where real-time decision-making depends on high-frequency, high-quality subsurface data, traditional cuttings workflows are often insufficient.

As horizontal drilling and hydraulic fracturing have revolutionized the development of unconventional resources, the industry has faced increasing demands for real-time, high-resolution subsurface characterization. In plays like the Eagle Ford (Figure 1), the complexity of stratigraphic architecture, combined with significant lateral and vertical variability in reservoir quality, mechanical properties, and organic content, necessitates detailed insights beyond what conventional tools can provide. The limitations of MWD and GR tools—especially in high ROP environments—include telemetry loss, insufficient resolution, gamma-ray washout, and tool calibration drift. While LWD and wireline tools can provide improved data quality, their deployment involves high costs, added rig time, and exposure to operational risks, particularly in extended-reach laterals or H₂S-prone environments.

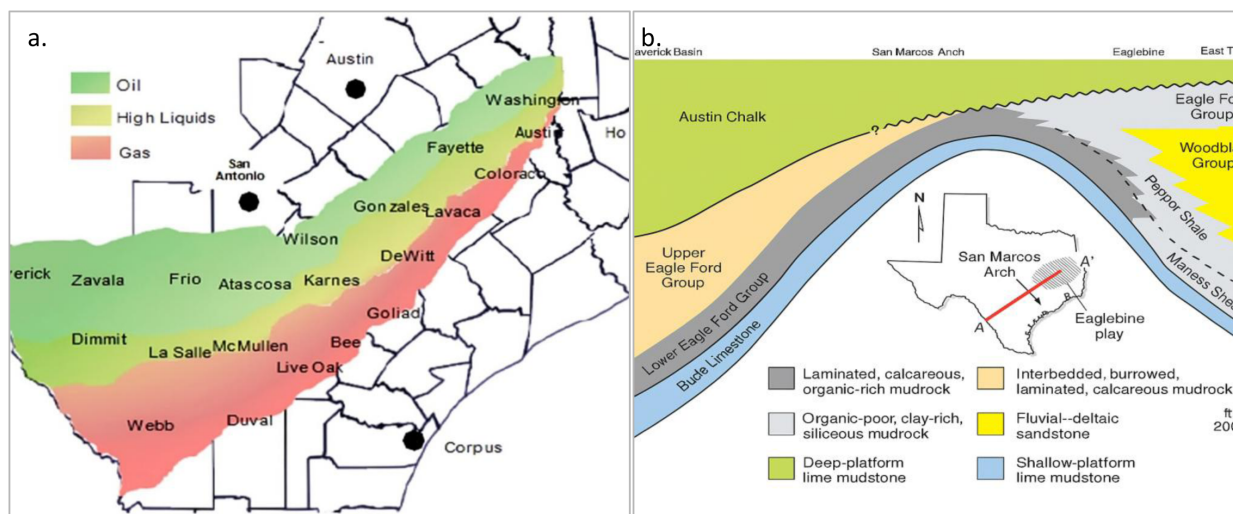


Figure 1—(a) Eagle Ford Shale distribution (From Bagci, 2017). (b) Regional cross-section A-A' from the Maverick Basin to the East Texas Basin (Hentz et al., 2014)

In response to these operational and geological challenges, Diversified's R&D team embarked on a multi-year initiative to transform drill cuttings from a low-value byproduct into a high-resolution, decision-driving dataset. In 2018, the team launched the Robologger™ project: a fully automated robotic sampling and analysis system designed for integration with rigsite operations. The technology suite enables consistent, high-frequency collection of drill cuttings directly at the shale shaker, eliminating the variability and limitations of manual workflows. The system includes an enclosed motive fluid loop to transport and clean samples, a real-time database for tracking sample metadata, and a direct interface with on-site analytical instruments.

Field trials and subsequent commercial deployment of this system began in the Eagle Ford Shale, a basin-centered, marine-deposited unconventional formation that spans over 400 miles from the Mexican border to East Texas. With an average thickness of 250 feet and comprising laminated calcareous mudstones, organic-rich shales, and volcanic ash layers, the Eagle Ford exhibits strong variability both laterally and vertically. Sub-seismic faulting and rapid facies transitions add to the challenge of maintaining wellbore placement within target intervals. Elevated bottomhole temperatures ranging from 200°F to 250°F, combined with complex pore pressure regimes, further complicate tool reliability and geosteering performance.

To address these challenges, the newly developed Robologger™ system was deployed in a variety of well configurations, including vertical pilot holes, deviated laterals, and multi-well pads. It was integrated with a portable benchtop XRF analyzer configured for major and trace element analysis and supplemented in certain cases with LIBS instruments to improve light element detection. Together, these tools enabled rapid turnaround of geochemical results, often within minutes of cuttings reaching the shaker. This allowed directional drillers and subsurface geologists to make near real-time steering decisions based on objective, quantitative data.

The introduction of this integrated workflow also introduced a significant advancement in health, safety, and environmental (HSE) practices. By removing personnel from direct cuttings handling in potentially dangerous areas and reducing reliance on manual sample processing, the system contributed to a safer working environment. Additionally, the automation helped lower overall sampling and analysis costs, as it reduced labor requirements and minimized rework caused by sampling errors or inconsistencies.

Methodology

This study employs a fully integrated and field-tested methodology that combines robotic cuttings acquisition, real-time elemental analysis, and advanced geochemical modeling to support formation evaluation, geosteering, and drilling optimization in unconventional reservoirs. The workflow was developed in-house by Diversified and refined during a two-year deployment across more than 80 horizontal wells in the Eagle Ford Shale.

Robotic Cuttings Acquisition System

At the core of the system is the Robologger™, a fully autonomous, programmable robotic sampling unit installed directly at the shale shaker's discharge (Figure 2). The device is engineered to collect drill cuttings at high frequency—typically at 5-foot depth intervals—even during high-rate drilling operations exceeding 400 feet per hour. Unlike manual sampling, which is labor-intensive, slow, and susceptible to inconsistency, the Robologger™ standardizes the cuttings retrieval process, eliminating human bias and reducing the likelihood of sample mislabeling or contamination.



Figure 2—Images of the RoboLogger™ system (Figures 2a and 2b) illustrate its ability to collect drill cuttings at fixed 5-foot intervals within an on-site laboratory at the rigsite.

The system utilizes a motive fluid loop—a closed, pressurized circuit—to transfer the collected samples from the rig floor to a modular field laboratory. During this transfer, the samples are subjected to mechanical agitation and fluid flushing to remove excess drilling mud and surface contaminants. This step is critical in preventing chemical interference from drilling additives such as barite or lignosulfonates, which can compromise geochemical data quality. Bypass valves, flushing chambers, and backflow preventers are embedded in the system to maintain flow assurance and sample integrity.

Each sample is assigned a unique identification code linked to metadata such as measured depth, bit depth, ROP, rotary torque, weight on bit (WOB), mud weight, and circulation rate. This digital traceability enables seamless data integration and allows for retrospective quality checks during post-well analysis.

Sample Preparation and Laboratory Setup

Upon arrival at the lab, samples are dried at 110°C, homogenized using a mechanical mill, and pressed into uniform pellets using a hydraulic press. This sample preparation protocol minimizes matrix effects, enhances signal stability, and ensures repeatable results across instruments and operators. Strict cleaning protocols and sample sequencing rules are followed to prevent cross-contamination between consecutive runs. Blank runs and replicate samples are inserted every 15–20 samples as part of the quality control program.

Sample throughput is optimized for real-time turnaround. A trained technician can process and analyze each sample in under 10 minutes, which enables near-continuous subsurface profiling during active drilling.

On-Site Geochemical Analysis

Elemental analysis is performed using a portable, field-deployable X-ray fluorescence (XRF) instrument configured to detect 11 major oxides (SiO_2 , Al_2O_3 , CaO , MgO , Fe_2O_3 , K_2O , TiO_2 , Na_2O , MnO , P_2O_5 , and SO_3) and 32 trace elements (including Sr, Zr, Ni, Cr, Rb, V, Mo, Ba, Zn, and U). Calibration is performed using matrix-matched standards and certified reference materials (CRMs) sourced from the USGS and NIST. Instrument stability is monitored through repeated measurement of internal standards at regular intervals throughout each shift.

In intervals of interest (e.g., organic-rich zones or carbonate transitions), Laser-Induced Breakdown Spectroscopy (LIBS) is also utilized to complement the XRF dataset. LIBS is particularly effective for detecting light elements such as hydrogen, lithium, boron, and carbon, which are not detectable by standard XRF. LIBS data are analyzed using multivariate regression and chemometric models to convert emission intensities into semi-quantitative elemental concentrations.

All geochemical data are processed through a proprietary software interface that conducts on-the-fly statistical checks, depth alignment, and unit conversion. Anomalous readings—such as spikes caused by contamination or sensor noise—are automatically flagged for technician review.

Elemental Gamma Ray (EGR) Modeling

To replace or supplement downhole gamma ray logs, the system generates a surface-derived elemental gamma ray (EGR) profile. This profile is calculated by converting XRF-derived concentrations of K, Th, and U into equivalent API gamma ray units. The model was calibrated using a historical dataset of over 200 wells with co-acquired XRF and MWD gamma logs, enabling the derivation of regression coefficients specific to the Eagle Ford's mineralogical context.

The API conversion formula includes matrix correction factors and scaling coefficients:

$$\text{EGR(API)} = a \cdot [\text{K}] + b \cdot [\text{Th}] + c \cdot [\text{U}] + e$$

Where:

- [K], [Th], and [U] are elemental concentrations in % or ppm;
- a, b, c, and e are empirically derived constants.

This EGR curve is plotted in real time alongside the drilling log, enabling correlation with GR responses from MWD or LWD tools. In cases of telemetry loss, tool shock, or sensor dropout, the EGR log provides a backup GR signal that can be used to maintain geosteering continuity.

Quality Control and Contamination Mitigation

Given the potential for drilling mud additives to interfere with geochemical analysis, the system includes a barium contamination flagging mechanism. Ba levels exceeding 10,000 ppm indicate probable barite carryover, which can distort the trace element dataset. Such samples are excluded from high-precision analyses or subjected to corrective algorithms based on local regression models.

Internal standards, duplicates, blanks, and CRMs are analyzed at defined intervals to ensure analytical fidelity. Daily instrument drift checks are conducted, and any shifts beyond the tolerance threshold prompt recalibration.

Stratigraphic and Geochemical Interpretation

Geochemical fingerprints are interpreted using elemental ratios known to indicate depositional environment, redox state, provenance, and mineralogical character. These include:

- Ni/Mo and V/Ni – proxies for anoxic and euxinic conditions (Figure 4,5)
- Rb/Zr and Rb/Sr – indicators of clay content and sediment input
- SiO₂/Al₂O₃ and CaO/MgO – discriminators of siliciclastic vs. carbonate lithologies
- TiO₂/Nb and Zr/Nb – flags for volcanogenic detritus or ash beds
- K₂O/SiO₂ – proxy for feldspar content and brittleness potential

Multivariate statistical methods—including principal component analysis (PCA), hierarchical cluster analysis (HCA), and k-means clustering—are applied to the geochemical data to define chemostratigraphic units, which are then cross-referenced with lithofacies interpretations, gamma-ray curves, and drilling performance metrics.

These data-driven chemostratigraphic units provide higher-resolution formation boundaries than conventional log-based picks and allow for cross-well correlation in areas where structure, faulting, or lateral facies transitions obscure traditional stratigraphy.

Results and Discussion

The integrated workflow—combining robotic sampling, real-time geochemical analysis, and surface-derived gamma ray modeling—was deployed across 80 wells spanning the southwest to northeast structural corridor of the Eagle Ford Shale. These wells represented a diverse subset of geologic settings, drilling objectives, and operational constraints, making them an ideal test bed for validating system performance, data quality, and impact on drilling outcomes.

Sampling Density and Subsurface Resolution

Traditional cuttings sampling at the rig site typically occurs every 20–30 minutes, translating to sample intervals of 80 to 120 feet, depending on rate of penetration. In contrast, the RoboLogger™ system delivered samples at consistent 5-foot intervals, resulting in an average of 1,800 to 2,200 samples per well—a 20-fold increase in vertical resolution. This enhancement enabled high-resolution profiling of stratigraphic boundaries, lithologic variability, and geochemical gradients that would otherwise go undetected.

Fine-scale features—such as transitional carbonate/siliciclastic boundaries, thin volcanic ash beds (Figure 3), and minor redox fluctuations—were resolvable within this denser dataset. For instance, the base of the Upper Eagle Ford member, often ambiguous in GR logs due to low contrast with overlying lithologies, was clearly delineated in multiple wells using inflections in CaO/MgO and Si/Al ratios combined with V/Mo enrichment trends.

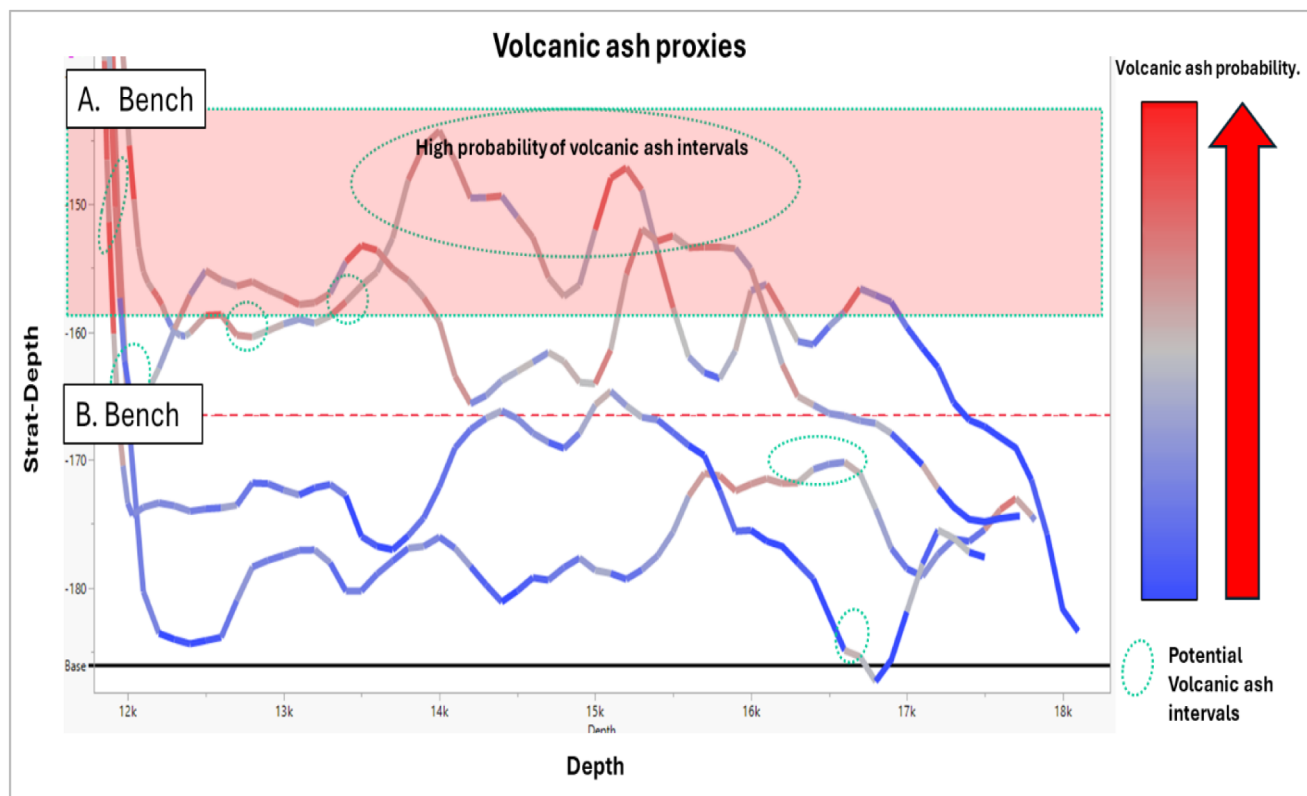


Figure 3—Volcanic ash proxies indicate a high probability of volcanic ash presence in Bench A compared to Bench B. The higher resolution sampling density allowed for a more detailed visualization of the data.

Gamma Ray Surrogate Validation and Use Cases

The elemental gamma ray (EGR) model was validated by comparing surface-derived GR logs to MWD gamma ray data across the 80-well dataset. In wells with high-quality MWD signals, the correlation coefficient between EGR and MWD logs exceeded **0.86** on average, with a maximum observed correlation of 0.94 in a clean carbonate-dominated lateral. The surface gamma log proved particularly valuable in intervals affected by MWD signal loss, washout artifacts, or GR calibration drift, offering a continuous and consistent alternative.

In one documented case, a 6,500-foot lateral encountered a persistent telemetry issue due to high vibration and lateral tortuosity. The EGR log—generated in real time from cuttings samples—was used to maintain stratigraphic targeting for over 4,000 feet of drilling. Post-well comparison with delayed LWD logs showed strong alignment in both baseline and gamma ray excursions, validating the utility of the surface proxy.

Operational Impact and Geosteering Optimization

The geochemical dataset played a direct role in operational decision-making throughout the campaign. By integrating XRF and LIBS-derived geochemical ratios into the geosteering console, directional drillers and geologists were able to interpret formation changes more confidently. Silica-rich zones, defined by elevated $\text{SiO}_2/\text{Al}_2\text{O}_3$ ratios and supported by declining CaO levels, were associated with **higher brittleness**, correlating with improved stimulation outcomes in nearby offset wells. These trends were used to adjust lateral trajectories in real time to maximize reservoir contact in brittle, frackable intervals.

Likewise, redox proxies (e.g., V/Cr, Ni/Mo, U/Th) helped delineate organic-rich intervals, which were targeted for enhanced landing zones. In multiple wells, the geochemical steering input prompted trajectory adjustments of 5–15 feet, ultimately improving exposure to high-TOC rock as later confirmed by wireline TOC and pyrolysis data.

In addition, Ti and Nb spikes, commonly associated with volcanic ash beds, were used as lateral correlation markers, aiding in stratigraphic tie-in across multi-well pads and mitigating miscorrelation in structurally deformed areas.

Contamination Identification and Quality Control

The inclusion of barium (Ba) monitoring in the analytical protocol enabled real-time identification of **mud contamination events**. In over 10% of wells, discrete intervals exhibited Ba concentrations >10,000 ppm, exceeding the defined threshold for potential barite interference. These flagged samples were excluded from chemostratigraphic models or were corrected using regression-based normalization against adjacent clean samples.

Replicate analysis of duplicate cuttings samples yielded average relative standard deviations (RSDs) of <3% for major oxides and <8% for trace elements, indicating robust repeatability and minimal sample preparation bias. Certified reference materials analyzed throughout the campaign consistently yielded recovery rates within $\pm 10\%$ of target values.

Brittleness and Mechanical Properties Mapping

An important outcome of the geochemical workflow was the derivation of a relative brittleness index, calculated using oxides associated with quartz and feldspar content (SiO_2 , K_2O) and carbonate dilution (CaO). This index showed strong correlation ($R^2 > 0.78$) with mechanical property logs, such as Young's modulus and Poisson's ratio, derived from wireline sonic data and sidewall core analysis.

Overlaying the brittleness profile with gamma ray and TOC proxies enabled multi-criteria landing zone optimization. In one example, a lateral initially planned to follow a GR-defined "clean" carbonate zone was instead geosteered into a geochemically derived brittle, organic-rich siliceous interval, resulting in a 15% increase in gas production compared to an adjacent well.

Chemostratigraphic Classification and Cross-Well Correlation

Using hierarchical clustering and PCA, chemostratigraphic packages were defined for the Eagle Ford interval, separating the Lower, Middle, and Upper members based on elemental fingerprints. Sub-units were identified within each member, including TOC-rich facies, silicified marker beds, and calcite-dominated interbeds, each defined by unique geochemical signatures.

This framework enabled more confident cross-well correlation in areas where GR logs were noisy, missing, or distorted by borehole conditions. It also provided a platform for regional stratigraphic mapping, particularly when combined with seismic facies and well log interpretations.

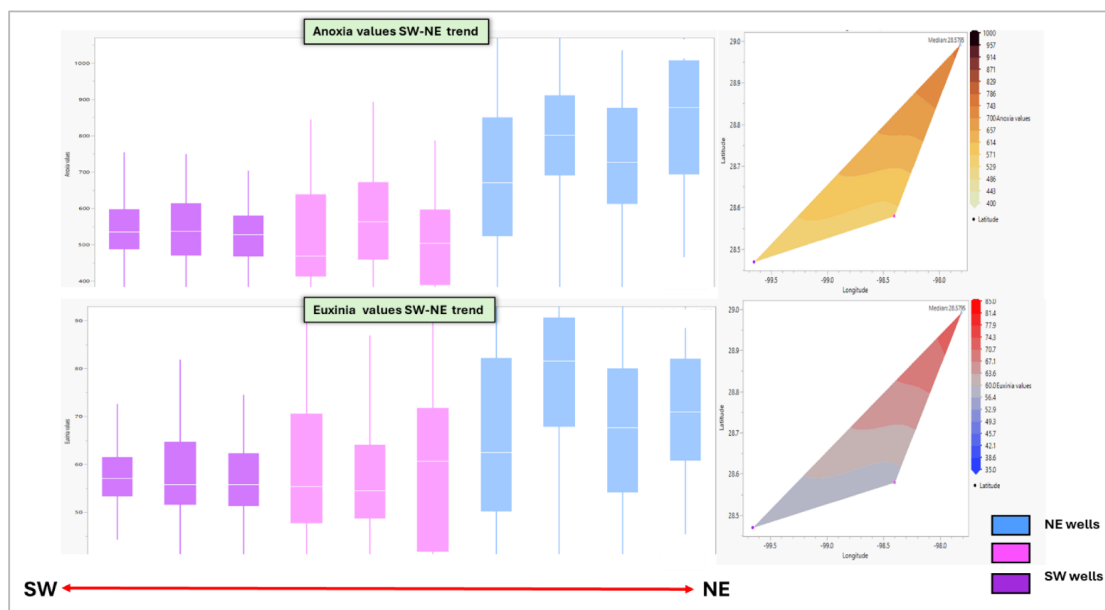


Figure 4—Distribution and maps of anoxia proxies along the SW–NE trend of the Eagle Ford Formation.

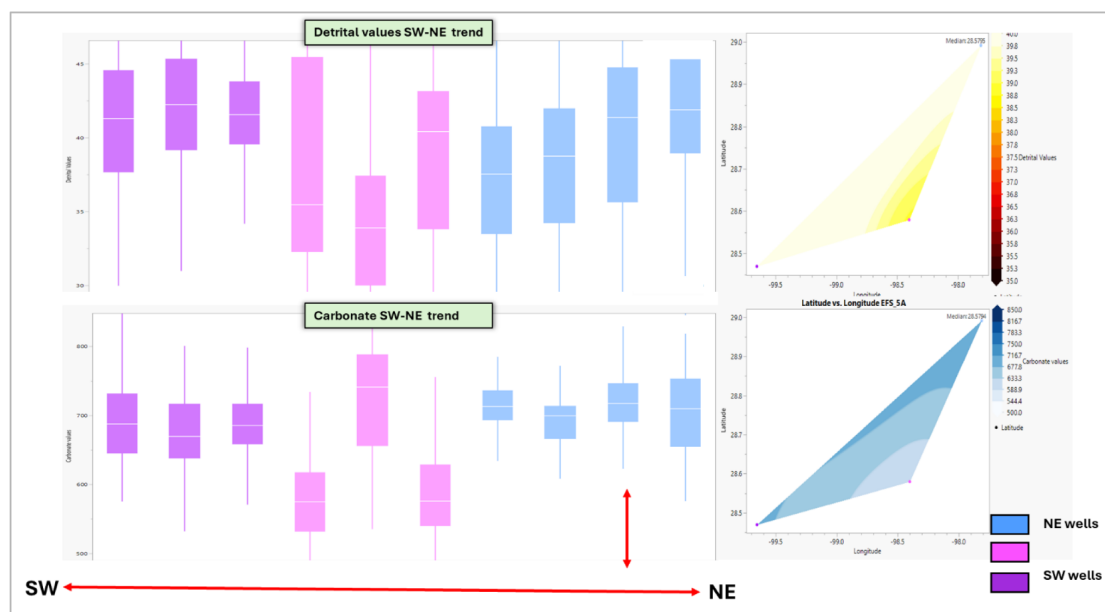


Figure 5—Distribution and maps of anoxia proxies along the SW–NE trend of the Eagle Ford Formation.

Conclusions

- The successful deployment of an integrated, automated workflow for drill cuttings acquisition and real-time geochemical analysis has demonstrated that high-density, high-resolution cuttings data can rival and, in many cases, supplement or replace conventional downhole measurements. In the geologically complex and operationally demanding environment of the Eagle Ford Shale, this methodology has not only improved formation evaluation fidelity but has also redefined the role of drill cuttings within the modern drilling workflow.
- The adoption of the RoboLogger™ system provided a step-change improvement in cuttings sampling frequency, enabling the acquisition of samples at 5-foot intervals—something previously unattainable using manual methods. This enhancement translated into significantly improved vertical resolution for chemostratigraphic interpretation, providing geologists and drillers with a detailed, continuous picture of the formation being penetrated. Unlike traditional gamma ray tools, which are susceptible to mechanical failure, calibration drift, and washout effects, the surface-derived Elemental Gamma Ray (EGR) log offered a robust, high-fidelity substitute derived from laboratory-grade elemental concentrations.
- The integrated use of portable XRF and LIBS instruments provided comprehensive major and trace elemental data across the wellbore. This data served multiple functions—supporting lithofacies identification, highlighting redox-sensitive organic-rich intervals, mapping carbonate versus siliciclastic content, and calculating proxies for brittleness, provenance, and depositional energy. Notably, the use of multi-element ratios and multivariate statistical clustering allowed for objective, reproducible chemostratigraphic classifications that were successfully correlated across dozens of wells and varied geological settings.
- From an operational perspective, the availability of high-frequency elemental data had measurable impacts on drilling and completions. Real-time geochemical steering enabled improved well placement in brittle, organic-rich facies, resulting in more effective fracture stimulation and enhanced reservoir contact. In cases where downhole tools failed—due to telemetry issues, vibration, or hole conditions—the surface-derived data stream served as a reliable fallback, reducing Non-Productive Time (NPT) and eliminating the need for costly tool redeployments.
- The system also delivered substantial HSE benefits by reducing the number of personnel exposed to hazardous conditions during manual sampling. The closed-loop nature of the sample transfer and analysis process ensured minimal physical contact with cuttings and eliminated errors associated with manual labeling, handling, and storage. From a cost perspective, the robotic system offered significant efficiencies by reducing labor requirements and increasing data reliability, ultimately improving the economics of drilling and logging operations.
- Furthermore, this workflow proved to be highly scalable and adaptable. While initially deployed in the Eagle Ford Shale, the methodology is transferable to other shale basins and conventional reservoirs where formation heterogeneity, downhole tool limitations, or budget constraints limit the effectiveness of traditional logging approaches. Plays such as the Bakken, Haynesville, Permian Basin (Wolfcamp/Bone Spring), and even international basins with less mature infrastructure stand to benefit from the implementation of similar surface-based geochemical monitoring systems.
- This study underscores the broader paradigm shift occurring in the upstream sector—a shift away from reliance on tool-centric, downhole measurements and toward surface-based, automated, data-rich solutions. In this context, drill cuttings are no longer viewed as waste or merely as a visual aid but as a continuous, rich, and accessible archive of subsurface geochemical information. Their real-time analysis represents a high-value opportunity for reducing cost, improving safety, and enhancing technical outcomes across the drilling and production lifecycle.

Key Takeaways

- 20x increase in sample resolution compared to manual collection enabled improved identification of formation transitions, brittle/ductile boundaries, and thin beds.
- Surface-derived gamma ray (EGR) logs correlated >85% with MWD GR data, offering reliable replacements during tool failure.
- Redox and brittleness proxies allowed landing zone refinement and informed completion design.
- Statistically derived chemostratigraphic units provided robust cross-well correlation in highly heterogeneous and faulted areas.
- Operational benefits included reduced NPT, lower exposure to HSE risks, and decreased reliance on expensive downhole tools.
- Scalability of the method makes it suitable for both mature and emerging basins with limited wireline coverage.

In conclusion, the integration of robotic sampling, real-time geochemical analysis, and elemental modeling offers a transformative and field-proven workflow that addresses both geological and operational challenges. It repositions drill cuttings as a primary source of high-resolution data for formation evaluation and wellbore optimization in the age of digital drilling and automation. The Eagle Ford case study serves as a blueprint for broader industry adoption of surface-based geoscience analytics in the development of unconventional resources.

References

1. Hentz, T.F., Ambrose, W.A., & Smith, D.C. (2014). Eaglebine play of the southwestern East Texas Basin: Stratigraphic and depositional framework of the Upper Cretaceous (Cenomanian–Turonian) Woodbine and Eagle Ford groups. *AAPG Bulletin*, **98**(12), 2551–2580. <https://doi.org/10.1306/08141413214>
2. Tonner, D., Swanson, A., Hollingshead, R., Hughes, S., Seacrest, S., McDaniel, B., & Leeper, J. (2018). *Automation provides unique insights of the rock record and subsurface through the delivery of a robotic sample collection and analysis device*. Presented at AAPG ACE 2018. Control ID: 2833118.
3. Tonner, D., Hughes, S., & Dix, M. (2012). *Wellsite geochemistry – New analytical tools used to evaluate unconventional reservoirs to assist in well construction and smart completions*. Presented at the SPWLA 53rd Annual Logging Symposium, June 16–20, 2012, Cartagena, Colombia.
4. Bagci, S. (2017). *Optimization of hydraulic fracturing and production enhancement: Case studies for US shale plays and tight sand reservoirs*. Presented at the 79th EAGE Conference and Exhibition – SPE EUROPEC, 12–15 June 2017, Paris, France. <https://doi.org/10.3997/2214-4609.201701474>